

Model V1 — AR(1) with Seasonality

Likelihood

$$y_{i,1} \sim \mathcal{N}(\mu_{i,1}, \tau_i^{-1}) \quad (1)$$

$$y_{i,t} \sim \mathcal{N}(\mu_{i,t} + \phi(y_{i,t-1} - \mu_{i,t-1}), \tau_i^{-1}), \quad t > 1 \quad (2)$$

Mean Function

$$\mu_{i,t} = \alpha_i + \beta_i \cos\left(\frac{2\pi t}{52}\right) + \gamma_i \sin\left(\frac{2\pi t}{52}\right) \quad (3)$$

Priors

$$\alpha_i, \beta_i, \gamma_i \sim \mathcal{N}(0, 0.001) \quad (4)$$

$$\tau_i \sim \Gamma(0.001, 0.001) \quad (5)$$

$$\phi \sim \text{Uniform}(-1, 1) \quad (6)$$